Qualifications for Software Developers: Education and Skills

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Software developers serve an important function for their employers by creating necessary internal and external programs. Understanding the varied qualifications necessary for this role can help you decide if this is the right career for you.

In this article, we discuss the qualifications for software developers, skills you can learn and how experience can affect getting jobs for this role.

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What is a software developer?

Software developers are professionals who create computer programs. Sometimes, the industry also refers to these people as computer programmers, software engineers, programmers or coders. Typically, they create computer software, though some may have duties that require them to create hardware for computers too. They often have extensive knowledge of programming languages and coding, which allows them to create programs that execute specific tasks. These tasks are typically determined by the needs and requests of clients who a software developer communicates with.

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Qualifications for software developers

Software developers often have several qualifications to be successful in their roles across multiple industries. Below are descriptions of the qualifications they may have:

Education

When you search for a role as a software developer, the job description may list a certain level of education. The most common education level for these employees is a bachelor's degree. Typically, the subjects you can pursue include:

- Software development
- Computer science
- Information technology
- Computer engineering
- Programming
- Coding

Some roles may also require applicants to have a master's degree. Typically, these are in the same subjects as the bachelor's degree. Usually, earning a master's degree can increase the average salary you make since you have more information and skills you can use while performing the duties of the role. While a master's degree isn't a requirement to become a software developer, it can definitely help you find ideal roles in the industry.

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Soft skills

Soft skills are those you can transfer between any roles you may have. They often represent qualities you have and ways you may interact with others that make you successful in your role. Below are the soft skills software developers can have to excel in their roles:

- **Communication:** Software developers often work with multiple other employees to finish the projects they have. They may also communicate with clients directly, so communication is critical to creating successful products.
- **Empathy:** Empathy is the ability to understand the emotional state of other people. This is helpful for software developers who work directly with clients and those who work with other employees at the same company because it allows them to relate to these other people.
- **Self-awareness:** Self-awareness is important for software developers because it allows them to find errors they may make in coding and programming more easily. It can also help them understand their strengths and use those to maximum effect while working on projects.
- **Patience:** Patience is important for software developers because they may work on projects that need a lot of work to be successful. On long projects, it may be easy to get frustrated as the project develops slowly, so patience can minimize frustration.
- **Open-mindedness:** Open-mindedness allows software developers to accept new ideas and incorporate them into projects. This means you can develop new practices and grow your skills throughout your career.
- **Problem-solving:** Software developers are often solving challenges for clients so understanding the process to do so is important for success.
- **Time management:** Many software developers work on multiple projects at the same time so it's important to manage time effectively so you can meet deadlines successfully.
- **Accountability:** Accountability is a valuable skill for software developers because it can show they understand what tasks they're responsible for and ensure they complete them to a high standard every time.

Related: Software Engineer vs. Developer: What's the Difference?

Hard skills

Hard skills are the ones you can develop through education, certification and experience in a role. They often represent job-specific skills you have. Below are the hard skills software developers may have:

- **Cloud security:** These are the processes that address internal and external information security within IT and computer systems.
- **DevOps:** These are the processes that improve development times and integrate high-quality systems.
- **Information technology (IT) automation:** These are software and other systems that can replace manual applications for repeatable processes.
- **Full stack development:** This is the process to create both front-end processes and back-end processes for programs from start to finish.
- **Programming languages:** These are the languages you can use to create programs and code to run specific processes. There are several programming languages and what you use can change based on your needs, the needs of the clients and the procedures of the company.
- **Testing procedures:** This is how you test programs and software to ensure it runs properly. It may also include stress testing to learn how well a program responds to unusual circumstances.
- **Operating systems:** These are the systems computers use to operate effectively. There are several systems and creating programs that work in all of them is a valuable skill for software developers.

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Certifications

Many software developers get certifications to show they're experts in specific processes. The certifications you pursue may also vary based on the role you have. Below are some common certifications for software developers:

- Certified Secure Software Lifecycle Professional
- Certified Software Development Professional
- Certified Software Engineer
- CIW Web Development Professional
- AWS Certified Developer Certification

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Is experience a qualification?

While experience as a software developer is preferable to some employers, it's not a typical requirement. Even so, getting experience in other roles or through freelance work can help you develop important skills that make you successful in a full-time role with a company. For

example, having several coding projects you've already completed can help you explain your specific skills to hiring managers during an interview.



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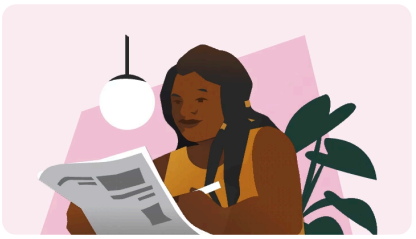
Jenn, a career coach, provides a look at the interviewing process and shares tips on how to position yourself for success at every step.

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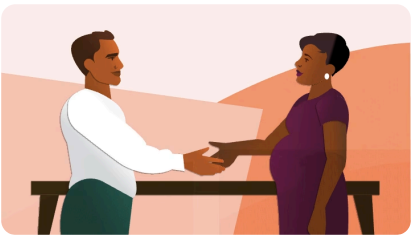
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